

MODULE 16 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Choosing Storage

Object, block, file or archive. The access pattern decides before any product does.

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Decide the **category** before the product.
Object, block, file, or archive is a question
about how the data is reached.

The four categories

CATEGORY	REACHED BY	SERVICE
Object	HTTP GET by key, whole objects	Amazon S3
Block	A disk, attached to one instance	Amazon EBS, or instance store
File	A mounted share, many clients	Amazon EFS, Amazon FSx
Archive	Restored before use, or slow reads	S3 Glacier classes

ON THE EXAM

Many clients writing to the same files at once eliminates EBS immediately, whatever the performance requirement says.

The S3 storage classes

CLASS	CHOOSE IT WHEN	WATCH FOR
Standard	Frequent access, unknown pattern	Costs most per GB
Intelligent-Tiering	Access pattern varies or is unknown	Small monitoring fee per object
Standard-IA	Infrequent but immediate	30-day minimum, retrieval fee
One Zone-IA	Infrequent and reproducible	One AZ. Losing it must be acceptable
Glacier Instant Retrieval	Archive, but read in milliseconds	90-day minimum
Glacier Deep Archive	Cold, hours to restore acceptable	180-day minimum

ON THE EXAM

Glacier Flexible Retrieval sits between the last two, restoring in minutes to hours. Instant Retrieval is the one people forget exists.

The S3 charges that surprise people

- ◆ **Minimum storage durations:** 30 days for the IA classes, 90 for Glacier, 180 for Deep Archive. Delete early and you still pay.
- ◆ **128 KB minimum billable object size** in the IA classes. A million tiny objects costs as if each were 128 KB.
- ◆ **Retrieval fees** apply to IA and Glacier. Frequently read data in a cheap class costs more than Standard.
- ◆ **Intelligent-Tiering has no retrieval fee**, which is why it beats a badly guessed lifecycle policy.

ON THE EXAM

A lifecycle rule moving small, frequently read objects to IA can raise the bill. Three separate charges make that happen.

EBS volume types

- ◆ **gp3 is the default.** 3,000 IOPS and 125 MiB/s baseline, provisioned independently of size.
- ◆ **gp3 now reaches 64 TiB, 80,000 IOPS and 2,000 MiB/s** since late 2025. It was 16 TiB and 16,000 IOPS.
- ◆ **io2 Block Express** for sub-millisecond latency, 99.999% durability, Multi-Attach, or beyond 80,000 IOPS.
- ◆ **st1 and sc1** are HDD, for large sequential throughput. Neither can be a boot volume.

ON THE EXAM

The old rule that anything above 16,000 IOPS needs io2 is now wrong by a factor of five. Study material still teaches it.

gp3 or io2 Block Express?

GP3

The default, now far bigger

- Up to 80,000 IOPS, 2,000 MiB/s
- Up to 64 TiB per volume
- Cheaper per GB and per IOPS
- Single-digit ms latency

IO2 BLOCK EXPRESS

For what gp3 cannot meet

- Up to 256,000 IOPS
- Sub-millisecond latency
- 99.999% durability
- Multi-Attach across instances

THE TIEBREAKER

Four reasons remain for io2: latency below a millisecond, five-nines durability, Multi-Attach, or above 80,000 IOPS. Nothing else.

Amazon EFS

- ◆ **NFS for Linux**, mounted by thousands of clients across Availability Zones at once. It grows and shrinks automatically.
- ◆ **Standard** spans multiple AZs. **One Zone** costs far less and dies with its zone.
- ◆ **Elastic throughput** scales with demand and is the sensible default. Provisioned suits steady high throughput.
- ◆ Lifecycle management moves untouched files to infrequent access and archive tiers automatically.

ON THE EXAM

Many Linux instances writing to shared files is EFS. Windows or SMB in the same sentence means FSx instead.

The FSx family

FSX FOR	GIVES YOU	REACH FOR IT WHEN
Windows File Server	SMB, with Active Directory	Windows apps, home directories, AD auth
Lustre	Massive parallel throughput	HPC, ML training, S3-linked scratch
NetApp ONTAP	NFS, SMB and iSCSI together	Migrating an existing ONTAP estate
OpenZFS	NFS with ZFS snapshots	Linux workloads wanting ZFS features

ON THE EXAM

SMB plus Active Directory is FSx for Windows. High-throughput compute against S3 data is Lustre. Those two carry most questions.

Storage Gateway

- ◆ **S3 File Gateway** presents NFS or SMB on-premises, storing files as objects in S3.
- ◆ **Volume Gateway, cached** keeps the full dataset in S3 and a hot subset locally. **Stored** keeps it all locally and backs up to S3.
- ◆ **Tape Gateway** presents a virtual tape library, so existing backup software writes to S3 and Glacier unchanged.
- ◆ **FSx File Gateway** gives on-premises low-latency access to an FSx for Windows file system.

ON THE EXAM

Cached keeps the primary copy in the cloud. Stored keeps the primary copy on-premises. That is the whole distinction.

Reading the scenario

THE SCENARIO SAYS

SMB share with Active Directory authentication

Thousands of Linux instances sharing files

Archive, but must be readable in milliseconds

Access pattern is unknown or changing

Local capacity exhausted, keep hot data fast

Above 80,000 IOPS, or sub-millisecond

IT IS POINTING AT

FSx for Windows File Server

Amazon EFS

S3 Glacier Instant Retrieval

S3 Intelligent-Tiering

Volume Gateway, cached mode

io2 Block Express

ON THE EXAM

Protocol and access pattern decide most storage questions. Performance numbers usually confirm a choice rather than making it.

Meridian Freight is sizing storage for a PostgreSQL database on EC2. It needs about 45,000 IOPS sustained, 8 TiB of capacity, and single-digit millisecond latency. There is no clustering requirement.

An architect has specified io2 Block Express, citing that gp3 caps at 16,000 IOPS.

What should the solutions architect advise?

- ☐ A Proceed with io2 Block Express as specified
- ☐ B Use gp3, which now supports up to 80,000 IOPS and 64 TiB
- ☐ C Stripe four gp3 volumes in RAID 0 to reach the required IOPS
- ☐ D Use st1 volumes, since databases are throughput-bound

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

io2 as specified
Would work, and costs more per GB and per IOPS for requirements gp3 now meets. The stated reason for choosing it is out of date.
- B

gp3
45,000 IOPS and 8 TiB both sit inside gp3's current limits, and single-digit millisecond latency is what gp3 delivers.
- C

RAID 0 across four volumes
The workaround from before the limits rose. It adds complexity and a wider failure surface for no benefit now.
- D

st1
HDD, capped near 500 IOPS. It is for sequential throughput and cannot serve a transactional database.

THE SIGNAL

The architect's reasoning was correct until September 2025. Check the numbers rather than the rule you remember.

A company must retain seven years of medical imaging for compliance. Files are almost never read, but when a clinician requests one it must be available within milliseconds, not hours.

Storage cost is the primary concern after that requirement.

Which storage class should the architect choose?

- ☐ A S3 Glacier Deep Archive, with expedited retrieval
- ☐ B S3 Glacier Instant Retrieval
- ☐ C S3 Standard-IA
- ☐ D S3 Intelligent-Tiering with the Deep Archive Access tier enabled

◆ PAUSE AND DECIDE

Which storage class should the architect choose?

A Deep Archive

Cheapest per GB and restores in hours. Expedited retrieval is not offered for Deep Archive, and the requirement says milliseconds.

C Standard-IA

Also millisecond access, and costs considerably more per GB than Glacier Instant Retrieval for data read this rarely.

B Glacier Instant Retrieval

Built for exactly this: archive pricing with millisecond access. Rarely read, immediately needed, cost-sensitive.

D Intelligent-Tiering with Deep Archive

Objects reaching the Deep Archive tier take hours to restore, which is the outcome the requirement forbids.

THE SIGNAL

Rarely read but instantly needed is the exact gap Glacier Instant Retrieval fills. Most people forget the class exists.

A finance team's Windows application needs a shared file system for 300 users, authenticating against the company's existing Active Directory, using SMB, with Windows ACLs preserved and shadow copies for user self-service restore. The application cannot be modified.

What should the solutions architect recommend?

- ☐ A Amazon EFS with an SMB gateway in front
- ☐ B Amazon FSx for Windows File Server, joined to the existing Active Directory
- ☐ C Amazon S3 with S3 File Gateway
- ☐ D An EC2 Windows file server on an io2 volume

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A EFS plus a gateway

EFS is NFS. Bolting on an SMB translation layer adds a component to maintain and still loses Windows ACL fidelity.

C S3 File Gateway

Presents a share backed by object storage. It is for extending on-premises capacity, not for hosting an AD-integrated share in AWS.

B FSx for Windows File Server

Native SMB, AD-joined, Windows ACLs preserved, and shadow copies built in. Every requirement is a native feature.

D Self-managed Windows file server

Works, and you now own patching, availability, backups and capacity for something AWS offers managed.

THE SIGNAL

SMB plus Active Directory is the pair. Four requirements here are FSx for Windows features and nothing else's.

A genomics team runs simulations across 500 EC2 instances. Each run reads a 40 TB reference dataset held in S3 and needs hundreds of gigabytes per second of aggregate read throughput. Results are written back to S3 and the scratch data is disposable. Runs happen twice a week and last about six hours.

What should the solutions architect recommend?

- ☐ A Amazon EFS with provisioned throughput
- ☐ B Amazon FSx for Lustre, linked to the S3 bucket, created for each run
- ☐ C Copy the dataset to an io2 Block Express volume on each instance
- ☐ D Read directly from S3 with a gateway endpoint

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

EFS provisioned
Genuinely shared and not built for hundreds of GB/s across 500 clients. Provisioning that throughput continuously would also be expensive.
- C

io2 per instance
500 copies of a 40 TB dataset, and EBS volumes are not shared. The copy alone would take longer than the run.
- B

FSx for Lustre linked to S3
Purpose-built for this. It presents the S3 data as a file system, delivers the throughput, and is deleted after each run.
- D

Direct from S3
Avoids a file system and gives object semantics and higher latency, which simulation code expecting POSIX files cannot use.

THE SIGNAL

Twice a week for six hours is the planted detail. It makes a transient file system created per run the economical answer.

An analytics platform writes about 40 million small JSON files a day, averaging 12 KB each. A lifecycle rule moves them to S3 Standard-IA after seven days, and a separate job deletes them after twenty-one days. The bill has risen since the rule was introduced. Files are occasionally re-read during the first two weeks.

Why has the cost increased, and what should change?

- ☐ A Standard-IA has a 30-day minimum and a 128 KB minimum billable size; keep them in Standard, or use Intelligent-Tiering
- ☐ B Move the objects to One Zone-IA instead to halve the cost
- ☐ C Move them to Glacier Instant Retrieval after seven days
- ☐ D Enable S3 Versioning to reduce duplicate storage

◆ PAUSE AND DECIDE

Why has the cost increased, and what should change?

- A

Minimum duration and minimum size

Deleted at 21 days but billed for 30, and 12 KB objects billed as 128 KB. Retrieval fees on re-reads make it worse still.
- B

One Zone-IA

Has the same 30-day minimum and the same 128 KB minimum. It reduces the per-GB rate on a bill inflated by both penalties.
- C

Glacier Instant Retrieval

Worse on both counts: a 90-day minimum against a 21-day lifetime, and higher retrieval fees on data still being read.
- D

Versioning

Keeps additional copies of objects, which increases storage. It has nothing to do with storage class economics.

THE SIGNAL

Three penalties compound here: early deletion, minimum object size, and retrieval fees. A cheaper class is not a cheaper bill.

A manufacturing site has 90 TB of design files on ageing on-premises NAS with only 20 TB of usable capacity left. Engineers need low-latency access to the roughly 8 TB they are actively working on, and the rest can tolerate slower access.

Replacing the NAS is not budgeted, and the applications use iSCSI.

What should the solutions architect recommend?

- A AWS Storage Gateway, Volume Gateway in cached mode
- B AWS Storage Gateway, Volume Gateway in stored mode
- C AWS DataSync to move everything to S3, and mount it with S3 File Gateway
- D Amazon FSx for OpenZFS accessed over Direct Connect

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Volume Gateway, cached
The full 90 TB lives in S3 while the active subset stays local. iSCSI is preserved and local capacity stops being the ceiling.
- B** Volume Gateway, stored
The exact inverse. Stored mode keeps every byte on-premises, which is the capacity problem they already have.
- C** DataSync plus File Gateway
File Gateway presents NFS or SMB. The applications use iSCSI, so they would need changing.
- D** FSx for OpenZFS over Direct Connect
Puts every read across the link, so the active 8 TB loses the low latency the scenario requires.

THE SIGNAL

Cached keeps the primary copy in the cloud with a hot local subset. Stored keeps the primary copy local. Capacity pressure means cached.

What to carry out of this module

- ◆ **Pick the category first.** Object, block, file or archive is a question about how the data is reached.
- ◆ **gp3 now reaches 80,000 IOPS and 64 TiB.** Only four things still require io2 Block Express.
- ◆ **Minimum durations and the 128 KB minimum** mean a cheaper class can produce a larger bill.
- ◆ SMB plus Active Directory is FSx for Windows. S3-linked high throughput is Lustre. Capacity pressure is cached Volume Gateway.

ON THE EXAM

Glacier Instant Retrieval is the class people forget: archive economics with millisecond access. It answers more questions than it should.